

DC-UPS-2CH

Quick user manual

for firmware 2026.09.02-v6 (build 190fc6b, 2026-09-04)

Dual-channel backup power supply for router, ONT and other compatible low-current network equipment.

Model	DC-UPS-2CH v2.1
Serial number	_____
Date of transfer	_____
Output ROUTER	_____ V / _____ A
Output ONT	_____ V / _____ A

Warning. Dangerous 230 V mains voltage is present inside the enclosure. The user must not open the enclosure while the mains or the battery is connected.

Keep this manual next to the device.

1. What the device does

DC-UPS-2CH automatically maintains power on two independent outputs. Under normal conditions the device is powered from the mains and keeps the battery in a charged state. When 230 V mains is lost the switch-over to the battery happens automatically, with no user action required.

Parameter	Value
Mains input	230 V AC, 50 Hz
Battery	12 V, 7 Ah, AGM/SLA
Float-charge voltage	13.8 V
Charge current limit	1.5 A
Battery early warning	11.5 V
Load disconnect (LVD)	11.0 V
Load re-enable threshold	12.5 V

Normal operation

Nothing needs to be switched over during a mains outage. The outputs keep running on the battery until its voltage reaches the protection threshold.

2. Daily use

2.1. Turning on

1. Confirm the required outputs are connected.
2. If the enclosure has separate mechanical output switches, set the required channels to the ON position.
3. Apply 230 V mains.
4. Wait for the connected equipment and Wi-Fi to come up.

Under normal operation no further manual control is required.

2.2. During a mains outage

The device automatically switches to battery. If a network path is available, an ntfy notification is sent. At battery voltage 11.5 V an early-warning notification is sent. At 11.0 V both managed outputs are shut off to protect the battery.

After protective shut-off the ESP32 enters emergency deep sleep. It wakes periodically and checks whether the mains has returned. When it does, the device automatically resumes normal operation.

Do not confuse with Shelf Sleep

After emergency deep sleep the device wakes automatically once the mains returns. In **Shelf Sleep** the return of 230 V does **not** wake the device.

3. Button and indication

3.1. Service button

Gesture	Result
1 short press	Send full status message to ntfy
2 short presses	Open DC-UPS-Setup access point for about 15 minutes
Hold for 10 s	Enter Shelf Sleep
Any press while in Shelf Sleep	Wake the device and perform a normal start-up

3.2. Button acknowledgement

- 2 green blinks — status successfully sent to ntfy;
- 3 red blinks — status could not be sent;
- short simultaneous flash of both LEDs — setup AP has been opened;
- 3 red blinks before shutdown — Shelf Sleep has been accepted.

3.3. Normal indication

Green LED	external power / mains present
Red LED	battery state, warning or emergency mode

While the device is booting and joining Wi-Fi, the green LED blinks at about 2 Hz. When the setup access point is active, both LEDs alternate slowly (green/red, 500 ms each side).

4. Shelf Sleep

Shelf Sleep is intended for transport and long-term storage, when both outputs and Wi-Fi must be completely off.

To power the device down for storage:

1. Hold the service button for about 10 seconds.
2. Wait for three red blinks.
3. Release the button.
4. Both outputs will switch off, Wi-Fi will go down, the ESP32 will enter deep sleep with no periodic wake-up.

To power the device on again: press the service button once.

Important

In Shelf Sleep the device wakes **only from a physical button press**. Connecting or restoring 230 V mains does not wake it.

5. Web panel and Wi-Fi

Once connected to the home network, the panel is available at:

`http://dc-ups.local/`

The IP address delivered through ntfy on every successful Wi-Fi connect can also be used directly.

The panel shows voltages, mains / output / Wi-Fi / internet status and the event log; lets the user manage ROUTER/ONT modes; perform channel restarts; and change Wi-Fi, ntfy and protection settings.

5.1. If the home Wi-Fi is not available

Double-press the service button quickly. An access point will appear:

DC-UPS-Setup

Default password: dc-ups-setup, unless it was changed during commissioning.

6. ntfy

The device sends notifications about important events on its own. In addition, one of the following commands may be posted to the same topic:

```
!ups    !ups status    !ups ping
```

The commands only **query** state and **do not control the outputs**. The reply includes battery voltage, 24 V rail, mains status, ROUTER/ONT, internet, uptime, RSSI and IP.

ntfy commands only work while the ESP32 is awake, connected to Wi-Fi and has internet access. In emergency deep sleep and in Shelf Sleep the Wi-Fi radio is off.

7. Connecting equipment

Before connecting a new router, ONT or other device, always verify:

- required output voltage;
- barrel plug polarity;
- permissible current of the corresponding output;
- the ROUTER / ONT labelling on the enclosure.

For the standard cylindrical DC connector of this unit, polarity must match the marking on the enclosure. Do not connect equipment when the required voltage or polarity is unknown.

8. Troubleshooting

Symptom	What to do
No power on the outputs	Check mains, mechanical switches, battery voltage and the per-channel mode in the panel
Web panel not reachable	Try <code>dc-ups.local</code> ; if it fails, double-press the button and connect to the setup AP
Outputs shut off after a mains outage	Normal LVD on low battery; wait for the mains to return
Device did not start after mains was restored	Shelf Sleep may be active — press the service button
3 red blinks after a short press	Status was not sent to ntfy; check Wi-Fi / internet / topic

If the fault repeats, do not open the device while it is energised and contact the manufacturer or system integrator.

9. Safety and maintenance

- do not operate the device with the enclosure open;
- do not allow water, condensation or airflow obstruction;
- do not short-circuit the battery terminals;
- do not exceed the output ratings printed on the label;
- the battery is a consumable part and loses capacity over time;
- if the autonomous runtime shortens noticeably, the battery should be checked and, if necessary, replaced according to the full service documentation.

Before opening the enclosure disconnect 230 V mains and the battery. Internal servicing is intended for trained personnel.

Quick reference

Normal operation: nothing to switch.

Mains lost: device works from the battery automatically.

Mains returned: device recovers from emergency deep sleep automatically.

1x button: status to ntfy.

2x button: setup AP.

Hold 10 s: Shelf Sleep.

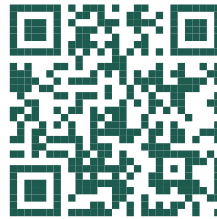
Wake from storage: press the button.

Panel: <http://dc-ups.local/>.

ntfy: !ups status or !ups ping.



Local web panel
(same LAN, mDNS)



Online documentation
(full passport, PDF)

Manufacturer contact: _____

Document date: 2026-09-04

Revision: User manual 2026.09.02-v6 / 190fc6b